

$$\begin{aligned}
 b) F(x, y, z) &= yz + wy' + wxz' + w'x'z \\
 &= yz \cdot 1 + wy' \cdot 1 + wxz' \cdot 1 + w'x'z \cdot 1 \quad (\text{Postulate 2(b)}) \\
 &= yz \cdot (x+x') + wy' \cdot (z+z') + wxz' \cdot (y+y') + w'x'z \cdot (y+y') \quad (\text{Postulate 5(a)}) \\
 &= yz \cdot x + yz \cdot x' + wy' \cdot z + wy' \cdot z' + wxz' \cdot y + wxz' \cdot y' + w'x'z \cdot y + w'x'z \cdot y' \quad (\text{Postulate 4(a)}) \\
 &= x \cdot yz + x' \cdot yz + wy'z + wy'z' + wxz' \cdot y + wxz' \cdot y' + w'x'z \cdot y + w'x'z \cdot y' \quad (\text{Postulate 3(b)}) \\
 &= (xy') \cdot z + (x'y') \cdot z + wy'z + wy'z' + wx \cdot (zy) + wx \cdot (zy') + w'x' \cdot (zy) + w'x' \cdot (zy') \quad (\text{Theorem 4(b)}) \\
 &= (xy') \cdot z + (x'y') \cdot z + wy'z + wy'z' + wx \cdot (yz) + wx \cdot (y'z') + w'x' \cdot (yz) + w'x' \cdot (y'z') \quad (\text{Theorem 4(b)}) \\
 &= xy'z + x'y'z + wy'z + wy'z' + wxz + wxz' + w'xy + w'x'y' \quad (\text{Theorem 4(b)}) \\
 &= xy'z \cdot 1 + x'y'z \cdot 1 + wy'z + wy'z' + wxz + wxz' + w'xy + w'x'y' \quad (\text{Postulate 2(b)}) \\
 &= xy'z \cdot (w+ w') + x'y'z \cdot (w+ w') + wy'z + wy'z' + wxz + wxz' + w'xy + w'x'y' \quad (\text{Postulate 5(a)}) \\
 &= xy'z \cdot w + xy'z \cdot w' + x'y'z \cdot w + x'y'z \cdot w' + wy'z + wy'z' + wxz + wxz' + w'xy + w'x'y' \quad (\text{Postulate 4(a)}) \\
 &= w \cdot xy'z + w' \cdot xy'z + w \cdot x'y'z + w' \cdot x'y'z + wy'z + wy'z' + wxz + wxz' + w'xy + w'x'y' \quad (\text{Postulate 3(b)})
 \end{aligned}$$

Removing the common minterms and placing them in order, we get:

$$\begin{aligned}
 &w'x'y'z + w'x'y'z + w'x'y'z + wx'y'z + wx'y'z + wx'y'z + wx'y'z \\
 &\Sigma(1, 3, 5, 9, 13, 14) \quad \text{Product of maxterms: } \Pi(0, 2, 4, 6, 8, 10, 11, 15)
 \end{aligned}$$

$$\begin{aligned}
 c) F(A, B, C, D) &= (A+B+C)(A+B')(A+C'D)(A+B+C'D)(B+C'D') \\
 &= ((A+B+C)+0)((A+B')+0)((A+C'D)+0)((A+B+C'D)+0)((B+C'D')+0) \quad (\text{Postulate 2(a)}) \\
 &= ((A+B+C)+D \cdot D')((A+B')+C \cdot C')((A+C'D)+B \cdot B') \quad (\text{Postulate 5(b)}) \\
 &= (A+B+C+D)(A+B'+C)(A+C'D+B)
 \end{aligned}$$